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Professional Summary:

- More than 10 years of IT experience with expertise in **Data Engineering, ETL/ELT Pipeline Development, LLM Data Infrastructure, Azure Cloud, and Business Intelligence**.
- Experienced in designing and building robust **ETL/ELT pipelines** for structured and unstructured data, including text scraping, cleaning, deduplication, and transformation workflows to support **LLM training and fine-tuning**.
- Hands-on experience with **Python, SQL, and NoSQL** for large-scale data management, with growing expertise in **LangChain, LlamaIndex, Hugging Face Transformers, and OpenAI API** for building LLM-powered applications.
- Proficient in implementing **Retrieval-Augmented Generation (RAG)** systems — connecting LLMs to company data using vector databases such as **Pinecone, Milvus, Weaviate, and Chroma** for embedding storage and semantic search.
- Strong background in cloud platforms (**Azure, AWS, GCP**), CI/CD pipelines, **Docker, and Kubernetes** for deploying and scaling AI/ML data infrastructure.
- Experience preparing high-quality datasets for fine-tuning LLM adapters (**LoRA**) and working with **PyTorch and TensorFlow** frameworks for model training and evaluation.
- Skilled in developing model evaluation frameworks to test for **accuracy, hallucination, and bias** in deployed LLM applications.
- Strong expertise in **Apache Spark** for high-throughput data processing, and familiarity with modern data stack tools including **Kafka** for real-time data streaming.
- Proficient in NLP techniques including **tokenization, word embeddings, and vector mathematics** to support AI/ML model development.
- Experienced building **APIs and internal web tools** for data annotation, curation, and model interaction.
- Strong SQL/T-SQL, data modelling, performance tuning, and troubleshooting skills across large-scale data systems including SQL Server, PostgreSQL, and Azure SQL.
- Good working experience in **Agile** methodology with strong collaboration across engineering, data science, and business teams.
- Strong experience working with **Team Foundation Server (TFS)** and **Git** for version control and CI/CD workflows.

Technical Skills:

AI / LLM Frameworks	LangChain, LlamaIndex, Hugging Face Transformers, OpenAI API, PyTorch, TensorFlow
Vector Databases	Pinecone, Milvus, Weaviate, Chroma
NLP & Embeddings	Tokenization, Word Embeddings, Vector Mathematics, RAG Systems, LoRA Fine-Tuning
Programming Languages	Python, T-SQL, PL/SQL, PostgreSQL, Shell Scripting, Basic C#, XML
Big Data & Streaming	Apache Spark, Apache Kafka, Azure Databricks (ADB), HD Insights, Cosmos DB
Cloud Platforms	Azure (ADF, ADLS, ADB), AWS, GCP
DevOps & Deployment	Docker, Kubernetes, CI/CD Pipelines, Git, TFS
Database	SQL Server 2012–2017, Azure SQL, PostgreSQL, Oracle 10g/11g/12c, NoSQL
ETL Tools	SSIS, Azure Data Factory (ADF), Alteryx, Apache Spark
BI & Reporting Tools	Power BI, SSRS, SSAS, Tableau, DAX, MDX

Applications	Visual Studio, SSMS, Power BI Desktop, SQL Server Profiler, Business Intelligence Studio
Version Control	Git, GitHub, Team Foundation Server (TFS)
Operating Systems	Windows, Linux, Mac OS

Professional Experience:

Client: Commonwealth of Massachusetts Sep 2020 – Till Date
Role: Senior Data Engineer / LLM Data Pipeline Engineer

Responsibilities:

- Designed and built robust **ETL/ELT pipelines** to ingest, clean, and transform large volumes of structured and **unstructured text data** from multiple source systems to support data-driven and AI/ML applications.
- Implemented **Retrieval-Augmented Generation (RAG)** workflows using **LangChain and LlamaIndex** to connect LLM models to enterprise datasets stored in vector databases, enabling semantic search and intelligent data retrieval.
- Built and maintained vector search solutions using **Pinecone and Chroma** to store and retrieve text embeddings for RAG systems, applying NLP techniques including **tokenization, word embeddings, and vector mathematics**.
- Developed **Python-based data preprocessing** workflows for cleaning, deduplication, and transformation of unstructured text data to prepare high-quality datasets for LLM fine-tuning using **LoRA adapters**.
- Implemented model evaluation frameworks to test LLM performance across **accuracy, hallucination detection, and bias** metrics, enabling continuous monitoring of deployed models.
- Created **APIs and internal tools** for data annotation, curation, and model interaction, enabling business users to efficiently label training data and interact with deployed LLMs.
- Leveraged **Azure Data Factory (ADF), ADLS, and Azure Databricks** to build high-throughput cloud data pipelines feeding AI/ML training workflows.
- Built **Power BI** dashboards and data models to visualise model performance metrics, data quality monitoring (DQM), and pipeline health using **DAX** queries and paginated reports.
- Developed and optimised **T-SQL stored procedures, CTEs, triggers**, and functions to consolidate data from disparate source systems with robust error handling and logging.
- Configured and maintained SQL Agent jobs for scheduled batch processing, monitoring pipeline consistency and performance across environments.
- Collaborated with data science and ML teams to translate business requirements into data infrastructure and model input specifications.
- Coordinated with offshore teams for day-to-day activities, updating senior management on project status and deliverables.

Environment: Python, LangChain, LlamaIndex, Pinecone, Chroma, Azure (ADF, ADLS, ADB), Docker, MS SQL Server 2016/2014, SSIS, SSRS, SSAS, Power BI, DAX, T-SQL, Git, GitHub, PowerShell, Visual Studio

Client: Equifax, St. Louis, MO July 2019 – Aug 2020
Role: Data Engineer / Business Intelligence Developer

Responsibilities:

- Built and maintained high-throughput **ETL/ELT pipelines** using **Python, Alteryx, and T-SQL** to extract, clean, deduplicate, and transform data from disparate sources including SQL Server, flat files, Excel, and XML.
- Developed data preprocessing workflows applying **NLP-adjacent text cleaning** and transformation techniques to prepare structured datasets for analytics and downstream ML model consumption.
- Implemented metadata-driven automated pipelines for incremental load processing, incorporating error handling, logging mechanisms, and master-child package configurations.
- Leveraged **Apache Spark** concepts and columnar storage strategies for performance tuning of large-scale analytical workloads and query optimisation.
- Created complex **stored procedures, user-defined functions, views, and indexing strategies** in SQL Server 2014/2017 to support high-performance data retrieval.

- Worked with **Semarchy xDM** for **Master Data Management (MDM)** including data integration, match and merge configurations, survivorship rules, and data quality validation.
- Designed data mapping documents for ETL solutions covering source, target, field name, and transformation logic for complex multi-system integrations.
- Developed dashboards and analytical reports in **Tableau and SSRS** to deliver business insights on key performance metrics.
- Managed **GitHub** for source code control and handled production deployments in an Agile team environment.

Environment: Python, SQL Server 2014/2017, Alteryx, T-SQL, SSRS, Tableau, GitHub, Semarchy xDM, SNOW (ServiceNow), Red Gate, Spotlight

Client: Michelin, Greenville, SC March 2018 – June 2019
Role: SQL Developer / Data Engineer

Responsibilities:

- Designed, developed, and tested ETL packages to extract, transform, and load data from multiple upstream sources (Feed Store, Cayman, MS Sales) into enterprise data warehouses using **SSIS and T-SQL**.
- Applied **data quality monitoring (DQM)** frameworks to assess accuracy of new data sources and preprocessing pipelines, building **Power BI** dashboards with slicers/filters for real-time DQM visibility.
- Processed large-scale Office Pricing and Exchange Rate datasets, designing automated Excel reports using Power Pivots and Power View for business intelligence consumption.
- Built and deployed tabular **SSAS** cubes and **Power BI** reports for critical KPIs including Monthly Active Users (MAU), Funnel Metrics, and Partner Margin analytics.
- Designed in-house **continuous integration and build automation** systems, ensuring reliable deployment of database and ETL objects via TFS version control.
- Created role-based access controls on cubes using geofencing for secure Power BI data delivery.
- Involved in **star schema** logical/physical data modelling with fact and dimension table design to support analytical workloads.

Environment: SQL Server R2/2012/2014/2016, SSIS, SSAS, SSRS, Power BI, Python, T-SQL, Visual Studio 2015, TFS, Excel

Client: Acrometis, King of Prussia, PA Jan 2017 – Feb 2018
Role: SQL Server / SSIS / SSRS Developer

Responsibilities:

- Built end-to-end **ETL pipelines** to extract, transform, and load source data into target data marts following defined data models and business rules, using **SSIS** with Task Factory components.
- Implemented column-level **data validation** and **ETL logging** at package and task level to track records processed, errors, and pipeline performance.
- Created complex **stored procedures, functions, indexes**, tables, views, and T-SQL code to support application data layers with full error handling.
- Migrated data from multiple sources to **SQL Server 2014** including XML, flat file, Excel, and OLE DB sources, and published to **Salesforce**.
- Developed parameterised, cascaded, drill-down, crosstab, and drill-through reports using **SSRS 2014** to deliver business intelligence to stakeholders.
- Managed deployments, job scheduling, and alerting for SSIS packages using SQL Agent jobs, ensuring reliable pipeline execution.
- Performed unit and system testing, troubleshooting, and bug fixing across production and QA environments.

Environment: MS SQL Server 2014/2016, SSIS, SSRS, SSAS, T-SQL, TFS, ASP.NET, VB.NET, Task Factory Tools, Windows

Client: Zeta Interactive PVT. LTD, India June 2015 – July 2016
Role: SQL Server / MS BI Developer

Responsibilities:

- Implemented **SSIS packages** for data exchange between disparate systems, migrating data from multiple source environments into SQL Server 2012/2014.
- Developed **ETL jobs** to extract, clean, transform, and load data from multiple sources into SQL Server tables, applying business transformation logic using **T-SQL stored procedures**.
- Generated operational and analytical reports using **SSRS 2012** including matrix, tabular, parametric, crosstab, drill-down, and ad-hoc reports via Report Builder.
- Automated database update functions to streamline salary and process-exception data ingestion, reducing manual effort through intelligent pipeline design.
- Created and maintained database objects including tables, indexes, stored procedures, and views, ensuring referential integrity and performance.
- Documented SSIS packages and report specifications, communicating requirements with business and end users.

Environment: SQL Server 2012/2014, SSIS, SSRS, T-SQL, SSMS, Business Intelligence Development Studio, TFS, Windows